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Data Analytics Week-3 Project Based on Advanced Excel, SQL, and R.

Week 3 Project List for a Data Analytics curriculum focusing on **Advanced Excel, SQL, and R**. These projects range across domains and skill levels, and are designed to reinforce data cleaning, querying, analysis, and visualization.

Here's a **Week 3 Project List** for a Data Analytics curriculum focusing on **Advanced Excel, SQL, and R**. These projects range across domains and skill levels, and are designed to reinforce data cleaning, querying, analysis, and visualization.

1. 🇮🇳 Week 3: Advanced Excel + SQL + R Projects (50 Projects)

#	Project Title	STEM Area	Description	Tools / Key Skills	Possible Datasets / Data Sources	Expected Deliverables
1	Environmental Sensor Data Monitoring	Science / Environmental Engineering	Collect air/water sensor readings, store in SQL, clean and detect anomalies in R, dashboard in Excel	SQL (data ingestion, time-series), R (anomaly detection, plotting), Excel (dashboard, pivot charts)	Govt data on air pollution, water quality; open sensor datasets	Excel dashboard showing pollutant trends + R report flagging anomalies
2	Astronomy Light Curve Analysis	Science / Physics	Use data of star brightness over time (light curves), store raw in SQL, analyze periodicity/variability in R,	SQL (time queries), R (Fourier transforms, time-series), Excel (visual summaries)	Kepler / TESS public data; astronomical databases	Report + Excel workbook of periodic stars, plots,

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			summarise in Excel			summary tables
3	Biomechanics / Motion Analysis	Engineering / Physics	Use motion capture / accelerometer data (e.g. gait), preprocess with SQL, analyze with R (signal processing), present key metrics in Excel	SQL (large data, window functions), R (filtering, spectral analysis), Excel dashboards	Sensor data from open datasets; human kinetics datasets	Excel dashboard of gait parameters; R scripts and statistical analysis
4	Crop Yield Prediction	Agriculture / Biology / Math	Combine soil, weather, fertilizer data, store & aggregate in SQL, build prediction model in R, Excel for decision support	SQL joins, aggregations; R regression / machine learning; Excel “what-if” scenarios	Satellite weather data, agricultural yield records	Prediction model + Excel tool for farmers: scenario / parameter adjustments
5	Renewable Energy Forecasting (Solar / Wind)	Engineering / Renewable Tech	Use historical solar/wind outputs plus weather data; SQL for data prep, R for forecasting time series, Excel dashboards	SQL time functions, R forecasting, Excel charting & sensitivity analysis	Public weather APIs; energy provider open data	Forecast reports + Excel dashboard with different scenario forecasts
6	Epidemiological Analysis of Disease Spread	Biology / Health / Math	Collect case, vaccination, demographic data; use SQL to manage, R to model spread, Excel to communicate visualization & trends	SQL filtering, joins; R models (SIR, logistic growth); Excel for infographics	WHO / CDC / Govt health data sets	R model outputs + Excel infographics & trend dashboards
7	Structural Engineering: Vibration Data Analysis	Engineering	From sensors on structures (bridges, buildings),	SQL for time-series data; R spectral	Data from structural health monitoring	Anomaly report + Excel dashboard

#	Project Title	STEM Area	Description	Tools / Key Skills	Possible Datasets / Data Sources	Expected Deliverables
			monitor vibrations, detect anomalies / fatigue using R; Excel for reporting, SQL for storage	analysis; Excel dashboards	; open sensor networks	for engineers
8	Oceanography : Sea Temperature & Salinity Trends	Environmental Science / Oceanography	Gather sea temp / salinity data across years; examine trends, seasonal cycles in R; dashboard in Excel	SQL, R seasonal decomposition, visualization; Excel summarization	NOAA / oceanographic public data sets	Trend plots, Excel dashboards, summary reports
9	Robotics Sensor Fusion / Error Correction	Robotics / Engineering	Work with multiple sensors (gyro, accelerometer, lidar), store calibration data in SQL, build sensor fusion algorithms in R, summary in Excel	SQL data prep; R filtering, Kalman filter; Excel overview & key metrics	Robotics datasets (e.g. UCI), measurement logs	R scripts + Excel report showing fusion performance, error rates
10	Meteorology: Storm / Rainfall Pattern Analysis	Meteorology / Data Science	SQL store rainfall data; R to detect patterns, extreme events; Excel to make maps/charts for public or policy use	SQL aggregates; R time series & spatial analysis; Excel visual reporting	National weather services, NOAA, open climate data	Extreme event detection + mapping + Excel dashboards
11	Physics Lab Data Automation & Analysis	Physics / STEM Education	Use measurements from lab experiments (e.g. motion, circuits), store raw in SQL, analyze (errors, regressions) in R, summary	SQL for storing & querying; R for regression / error analysis; Excel for lab reports	Student lab data or public physics lab datasets	Excel-formatted lab report template + R analyses

#	Project Title	STEM Area	Description	Tools / Key Skills	Possible Datasets / Data Sources	Expected Deliverables
			sheets & graphs in Excel			
1 2	Mathematics: Fractal / Chaotic System Visualizations	Mathematics / Theoretical STEM	Compute and store series or iterative map data in SQL; generate fractals / phase space plots in R; present visuals in Excel	SQL numeric data; R plotting; Excel embedding visuals & summaries	Synthetic data or known chaos system definitions	Visual plots, Excel workbook with interactive graphs
1 3	Neuroscience: Brain Signal / EEG Analysis	Biology / Neuroscience	EEG data: store high frequency time series in SQL, clean & extract features in R (e.g. power spectra), Excel to summarize for clinicians	SQL for large time-series; R for signal processing; Excel dashboards	Open EEG datasets (e.g. PhysioNet)	Feature tables, plots + Excel summary dashboards
1 4	Material Science: Stress-Strain Curve Analysis	Engineering / Material Science	Collect lab data of material testing; SQL for data storage; R for modeling yield, elasticity; Excel for summarization	SQL data handling; R curve fitting; Excel charts and comparison tables	Material testing datasets or lab experiments	Model fits + comparative charts in Excel
1 5	Chemical Reaction Rates Kinetics	Chemistry / STEM	Reaction rate data stored in SQL, R used to fit kinetic models, Excel for calculation sheets & visualization	SQL storage; R nonlinear modeling; Excel formula sheets	Chemistry lab data or published kinetic datasets	Rate constants, model residuals, Excel graphs
1 6	Genomics Data Analysis (Counts / Expression)	Biology / Bioinformatics	Count data (gene expression), store in SQL, normalize & cluster in R, Excel for high-level summaries & heatmaps	SQL large table handling; R bioinformatics (edgeR / limma), clustering; Excel heatmaps & summaries	Public gene expression datasets (GEO, ENA)	Clustered heatmaps, top genes, Excel summary sheets

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17	Environmental Impact of Urban Heat Islands	Environmental Science / Urban Engineering	Use city temperature, land cover data; SQL to combine sources; R to model heat island effect; Excel to present maps/trends	SQL joins, geospatial; R modeling, mapping; Excel for infographics	Remote sensing data, local weather stations	Heat maps, trend reports, Excel dashboards
18	Battery Performance Over Cycles	Engineering / Material Science / Energy Storage	Store discharge/charge cycle data in SQL; analyze capacity fade & internal resistance in R; report in Excel	SQL, R modeling, Excel summarization & charts	Lab battery test data, open battery datasets	Degradation curves, Excel tools to predict lifespan
19	Biomedical Imaging Data (e.g. MRI/CT) Summary Metrics	Biomedical Engineering	Extract metrics (volume, intensity) from image metadata; store summary in SQL; statistical comparisons in R; summary dashboard in Excel	SQL metadata extraction; R stats; Excel reporting	Public imaging study metadata	Comparative statistics, visuals + Excel summaries
20	Quantum Physics: Simulation Data Analysis	Physics / Theoretical STEM	Simulated quantum system data stored in SQL; analyze results in R; Excel to help visualize & compare parameter sweeps	SQL to store simulation outputs; R for analysis; Excel reporting	Simulation outputs, published computational physics data	Parameter sensitivity analysis, Excel reports
21	Math Modeling: Epidemiological SIR/SEIR Simulations	Mathematics / Health	Run simulations, store various parameter outputs in SQL; analyse outcomes in R;	SQL for scenario storage; R modeling; Excel scenario dashboards	Synthetic simulation data or historical outbreaks	Scenario comparison, Excel tools for model

#	Project Title	STEM Area	Description	Tools / Key Skills	Possible Datasets / Data Sources	Expected Deliverables
			Excel summary & comparisons			exploration
22	Engineering: Fault Detection in Mechanical Systems	Mechanical / Electrical Engineering	Sensor data of vibrations, temperature etc in SQL; R for detecting anomalies/faults; Excel dashboard for maintenance planning	SQL window stats; R anomaly detection; Excel alerting visuals	Machine sensors, industrial open datasets	Fault detection reports, Excel maintenance schedules
23	Robotics: Path Planning Performance Analysis	Computer Engineering / Robotics	Log path data; store in SQL; analyze traveled vs optimal paths in R; Excel summaries	SQL path data queries; R analysis; Excel performance tables	Robotics logs or simulated path planning data	Efficiency metrics, Excel report comparing algorithms
24	Ocean Plastic Pollution Mapping	Environmental Science / Marine Biology	Collect location data of plastic pollution; SQL to store; R spatial analysis; Excel to map & quantify burden	SQL geospatial; R mapping; Excel charts & maps	Citizen science data; mapping projects	Maps + numbers (tonnage, location) + Excel dashboards
25	Climate Change: Glacier / Ice Melt Monitoring	Earth Science / Environmental	Historical ice cover data; SQL + R trend & rate analysis; Excel visual reports	SQL historical datasets; R trend models; Excel charts & maps	Satellite imagery / climate data portals	Melting rates, trend forecasts & Excel overview
26	Fluid Dynamics: Flow Visualization & Analysis	Engineering / Physics	Experimental or simulated flow data stored in SQL; R to analyze vortices etc; Excel to summarize key parameters	SQL large dataset handling; R specialized analysis; Excel summaries	Simulations or experimental data from fluids lab	Parameter tables, visualizations & Excel summary
27	Plant Phenotyping / Growth Analysis	Biology / Agriculture	Measure plant growth under different treatments; SQL to store; R to analyze growth	SQL for experimental design; R curve fitting; Excel for visuals	Experimental datasets; public plant	Growth rate models, treatment comparison

#	Project Title	STEM Area	Description	Tools / Key Skills	Possible Datasets / Data Sources	Expected Deliverables
			curves; Excel charts for comparisons		growth studies	ns + Excel dashboard
28	Robotics: Drone Flight Data Analytics	Technology / Engineering	Analyze drone telemetry: altitude, battery, GPS; SQL storage; R to analyze performance; Excel for mission summaries	SQL time series & geospatial, R; Excel mapping & dashboards	Drone flight logs / simulated data	Flight performance metrics, Excel dashboards
29	Network Analysis in STEM Collaboration	Mathematics / Social Science	Study collaboration networks among STEM researchers; SQL to store edges/nodes; R to compute network metrics; Excel for summaries	SQL graph representation; R igraph / network packages; Excel summary	Publication / coauthor data sets	Network centralities, clusters, Excel reports
30	Biome Ecology: Species Distribution Modeling	Ecology / Biology	Presence/absence data; SQL to manage; R modeling (GLMs, MaxEnt); Excel maps & summaries	SQL spatial data; R statistical ecology; Excel visual summary	GBIF data or biodiversity portals	Species distribution maps, key variables + Excel summary
31	Materials: Thermal Conductivity vs Composition	Material Science / Chemistry	Data of material samples, composition vs thermal performance; SQL to store; R for modeling; Excel reporting	SQL for dataset; R regression / ML; Excel for comparison & charts	Lab sample datasets or published research	Predictive model + Excel tool to compare compositions
32	Fishery Stock Assessment	Marine Biology / Environment	Historical catch & survey data; SQL to store; R for stock	SQL time series; R population models; Excel dashboards	Fisheries survey data	Stock status outputs, Excel

#	Project Title	STEM Area	Description	Tools / Key Skills	Possible Datasets / Data Sources	Expected Deliverables
		ntal Science	modeling; Excel report for policy makers			policy report style
33	Seismology: Earthquake Event Analysis	Earth Sciences / Geophysics	Event catalogues, magnitude, depth etc; SQL to store; R for spatial & temporal clustering; Excel visuals	SQL queries; R clustering / spatial; Excel dashboards & maps	USGS or regional earthquake data	Patterns of quakes, risk zones + Excel summaries
34	Biomedical Signal: Heart Rate Variability	Medicine / Biology	Time series of heart beats; SQL store; R for HRV metrics; Excel summary dashboards	SQL time series; R signal processing; Excel charts	Open health datasets (PhysioNet)	HRV metrics, comparisons + Excel dashboards
35	Chemistry: pH / Concentration Titration Curves	Chemistry	Titration experiment data stored in SQL; R analyze equivalence points; Excel visualization & summary sheets	SQL for raw data; R fitting; Excel plots	Lab datasets or simulated data	Curve fits + summary tables, Excel visuals
36	Environmental Noise Pollution Mapping	Environmental Science / Public Health	Collect noise level readings spatially; SQL + R spatial interpolation; Excel maps & dashboards	SQL for geodata; R mapping/interpolation; Excel summary	Urban noise datasets or own collection	Noise map, reports & Excel dashboards
37	STEM Education: Student Performance in STEM Subjects	Education / Math	School data: grades, attendance; SQL to store; R for predictors; Excel reports for educators	SQL joins; R regression / descriptive; Excel dashboards	School or public exam datasets	Predictors of performance, Excel dashboards for teachers
38	Mathematics: Optimization of Engineering Designs	Engineering / Math	Data from designs (e.g. wing shapes, stress tests);	SQL data handling; R optimization;	Simulation or lab data	Optimal parameters + Excel

#	Project Title	STEM Area	Description	Tools / Key Skills	Possible Datasets / Data Sources	Expected Deliverables
			SQL to store; R to do optimization; Excel summary	Excel comparison charts		comparison tools
39	Hydrology: River Flow & Flood Risk Modeling	Earth / Environmental Engineering	River flow data, rainfall; SQL to store; R modeling; Excel maps & dashboards	SQL hydrological data; R models; Excel visuals	Hydrology datasets; gauge station data	Flood risk maps + Excel dashboard
40	Astrophysics: Exoplanet Transit Detection	Astronomy / Physics	Light curves, detect dips; SQL storage; R for signal detection; Excel summary & visuals	SQL Time series; R signal detection; Excel charts	Public exoplanet transit data	Transit detections + summary dashboards
41	Chemoinformatics: Drug Molecule Property Prediction	Chemistry / Computational Biology	Molecular descriptors; SQL storage; R for model building; Excel summaries	SQL data tables; R ML; Excel model comparison	Public chemical databases e.g. ChEMBL	Model outputs + Excel tool for property prediction
42	Engineering: Thermal Stress Analysis of Components	Mechanical / Aerospace Engineering	Temperature cycles data; stress strain; SQL storage; R modeling; Excel reports	SQL, R curve fitting; Excel comparison dashboards	Lab or simulation data	Component lifetime predictions + Excel dashboard
43	Meteorology: Wind Farm Layout Optimization	Renewable Energy / Engineering	Wind patterns; SQL for storing wind data; R for spatial modeling; Excel for cost vs efficiency tradeoffs	SQL spatial time series; R spatial models; Excel dashboards	Wind resource datasets	Optimal layout suggestions + Excel summary
44	Geology: Mineral Composition vs Hardness	Geology / Material Science	Mineral sample compositions; SQL storage; R modeling; Excel summary charts	SQL data tables; R regression; Excel visuals	Geological sample datasets	Correlations, model, Excel visuals

#	Project Title	STEM Area	Description	Tools / Key Skills	Possible Datasets / Data Sources	Expected Deliverables
45	Physics: Optics Interference / Diffraction Data	Physics / STEM	Collect diffraction pattern data; SQL store; R analyze fringe spacing; Excel visuals for lab reports	SQL data storing; R analysis; Excel plots	Lab experiment data or published datasets	Analysis of fringes + Excel summary report
46	Robotics: Battery & Energy Consumption Optimization	Engineering / Robotics	Robot power consumption vs load; SQL store; R to model, Excel for trade-off summaries	SQL monitoring logs; R modeling; Excel dashboards	Robot logs or simulated robotics data	Efficiency curves + Excel decision tool
47	Genetics: Heritability Estimation	Biology / Genetics	Family trait data; SQL storage; R statistical genetics; Excel summaries	SQL data; R mixed models; Excel summary	Genetic / pedigree datasets	Heritability estimates + Excel report
48	Aerodynamics: Airfoil Pressure & Flow Data	Aerospace / Engineering	Pressure/velocity data; SQL storage; R flow analysis; Excel visualizations	SQL, R, Excel plotting	Wind tunnel / CFD simulation data	Flow plots + Excel dashboards
49	Mathematics: Stochastic Processes Simulation & Analysis	Math / Probability	Simulate random walks / Poisson processes, store outcomes in SQL; analyze in R; Excel visual summary & comparisons	SQL for storage; R simulation & stats; Excel dashboards	Synthetic simulation data	Statistical summaries, comparison & Excel report
50	Biomedical: Drug Response Curve Analysis	Pharmacology / Biomedical Science	Dose-response data, SQL storage; R to fit models (e.g. sigmoidal), Excel for visual & decision making sheets	SQL data; R nonlinear fitting; Excel visualization & summary	Experimental or published dose-response datasets	EC50/IC50 values + Excel summary and plots

Playbook for for Advanced Excel + SQL + R Project

(Use below data As a Reference while making the above listed projects)

Absolutely! Here's a **comprehensive playbook** for executing an **Advanced Excel + SQL + R project** step-by-step. This guide is designed to help you **structure your workflow** efficiently, covering from data acquisition to analysis and reporting.

Playbook for Advanced Excel + SQL + R Project

Here's a **comprehensive playbook** structure you can follow for the **50 STEM-based Advanced Excel + SQL + R Projects**. This guide breaks down the process into **6 key phases** applicable to *any project* from the list, with detailed steps, tools, and examples. You can tailor this for individual projects.

Playbook: Advanced Excel + SQL + R for STEM Projects

1. Phase 1: Project Planning & Scoping

Step	Description
1.1	Select a Project Topic from the STEM list based on your interest (e.g., "Crop Yield Prediction" or "EEG Analysis").
1.2	Define Objectives – What problem are you solving? What insights or predictions are expected?
1.3	Identify Target Users – Engineers, scientists, policymakers, students, etc.
1.4	List Key Questions – e.g., "Which factors affect crop yield the most?" or "Can EEG signals detect early signs of epilepsy?"

2. Phase 2: Data Collection & Storage (SQL)

Step	Description
2.1	Find Data Sources – Use public STEM datasets (e.g., NOAA, PhysioNet, Kaggle, GBIF, NASA, etc.)
2.2	Import to SQL – Use tools like MySQL, PostgreSQL, SQLite, or cloud platforms like BigQuery or AWS RDS.

Step	Description
2.3	Design Schema – Normalize tables (e.g., <code>sensor_readings</code> , <code>locations</code> , <code>time</code> , <code>experiments</code>).
2.4	Data Cleaning via SQL: - Remove duplicates - Handle NULLs - Create indexes - Use date/time functions
2.5	Example SQL Query:

```
SELECT location, AVG(temperature) AS avg_temp
FROM weather_readings
WHERE date BETWEEN '2021-01-01' AND '2021-12-31'
GROUP BY location;
```

3. 🧠 Phase 3: Data Analysis & Modeling (R)

Step	Description
3.1	Connect R to SQL (via <code>RMySQL</code> , <code>RPostgreSQL</code> , or <code>DBI</code> packages):

```
library(DBI)
con <- dbConnect(RMySQL::MySQL(), dbname = "stem_project", host =
"localhost", user = "root")
data <- dbGetQuery(con, "SELECT * FROM sensor_readings")
```

| 3.2 | **Clean Data in R** using `dplyr`, `tidyr`, `lubridate` |

| 3.3 | **EDA (Exploratory Data Analysis)** using `ggplot2`, `skimr`, `summarytools` |

| 3.4 | **Apply STEM Models** depending on project type: - **Linear Regression** (e.g., material testing) - **Clustering** (e.g., gene expression) - **Time-Series (ARIMA, Prophet)** (e.g., solar energy) - **Signal Processing** (EEG, ECG) - **SIR/SEIR Models** (disease modeling) |

| 3.5 | **Sample R Modeling Code:**

```
model <- lm(yield ~ rainfall + soil_nitrogen, data = data)
summary(model)
```

4. 📊 Phase 4: Reporting & Visualization (Excel)

Step	Description
4.1	Export Cleaned Data or Model Outputs from R:

```
write.csv(final_data, "outputs.csv")
```

| 4.2 | **Import to Excel** – Use Power Query or Data → Import. |

| 4.3 | **Build Dashboards** with:

- PivotTables
- Slicers
- Conditional formatting
- Charts (line, bar, scatter)

- Sparklines
- What-if Analysis (Scenario Manager for simulation-based projects) |
- | 4.4 | **Templates to Include:**
- Executive Summary tab
- Charts Dashboard tab
- Data tab (imported from CSV or database)
- Metrics tab with KPIs |
- | 4.5 | **Sample KPI Dashboard Elements:**
- Total Yield Change
- Anomaly Count
- Predicted vs Actual
- Forecast Confidence

5. Phase 5: Deliverables & Presentation

Step	Description
5.1	Create a Project Folder Structure:

```

/Project_Name/
├── SQL/
├── R_Scripts/
├── Excel_Report/
├── Data/
└── Documentation/

```

| 5.2 | **Write a Brief Report** with:

- Objective
- Methodology
- Key Findings
- Screenshots of Excel Dashboard
- Interpretation of R Outputs |
- | 5.3 | **Prepare a Slide Deck (Optional)** for stakeholder presentation or academic review. |

6. Phase 6: Iteration & Enhancement

Step	Description
6.1	Collect Feedback from domain experts or users.
6.2	Enhance Model Accuracy – Add new variables, tune hyperparameters.
6.3	Automate Updates using:

- R scripts in RStudio or cron jobs
 - Excel Power Query auto-refresh
 - SQL scheduled tasks |
-

Templates You Can Create

Template Type	Purpose
Excel Dashboard	To present KPIs and visualizations
SQL Schema Script	To set up your database
R Script Template	Data connection → cleaning → modeling
Report Template (Word/PDF)	To summarize methodology, results
Presentation Slide Deck	For academic, client, or stakeholder delivery

Example Use Case Snapshot (Crop Yield Prediction)

Tool	Use
SQL	Store weather, crop yield, and soil data from 10 years
R	Clean and build a linear regression model
Excel	Interactive dashboard to forecast next year's yield with sliders for rainfall/fertilizer levels

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